

Power Monitor with PID Braking Control

Overview

This project implements an ESP32-based power monitoring system with PID-controlled braking. It measures voltage, current, and power using an INA226 sensor, displays readings on an OLED screen, and publishes data via MQTT. The system features automatic braking when voltage exceeds a threshold, controlled by a PID algorithm for precise regulation.

Hardware Components

- ESP32 development board
- INA226 current/voltage sensor
- SH1106G OLED display (128x64)
- Braking circuit connected to PWM outputs
- WiFi connection
- MQTT broker

Pin Configuration

Component	ESP32 Pin
INA226 SDA	18
INA226 SCL	19
OLED SDA	32
OLED SCL	33
Brake Pin 1	14
Brake Pin 2	25

Features

- Real-time monitoring of voltage, current, and power
- OLED display with clear readings
- WiFi connectivity
- MQTT data publishing in JSON format
- PID-controlled braking system
- Multiple operation modes (GEN, BRAKE, IDLE)

Software Dependencies

- Wire.h
- WiFi.h
- PubSubClient.h
- ArduinoJson.h
- Adafruit_GFX.h
- Adafruit_SH110X.h
- INA226.h
- PID_v1.h

Setup Instructions

1. Create a `secrets.h` file with your WiFi and MQTT credentials:

```
#define WIFI_SSID "your_wifi_ssid"
#define WIFI_PASS "your_wifi_password"
#define MQTT_BROKER "your_mqtt_broker_ip"
```
2. Connect hardware according to the pin configuration
3. Upload the code to your ESP32
4. Ensure your MQTT broker is running

MQTT Communication

- Published topic: `powermonitor/data` (contains voltage, current, power readings)
- Status topic: `powermonitor/status`
- Control topic: `powermonitor/setpoint` (to change braking threshold)

Operation

The system monitors power parameters and automatically activates braking when voltage exceeds the setpoint (default 8.0V). The PID controller adjusts PWM duty cycle to maintain voltage at the desired level.

Troubleshooting

- The system will retry INA226 initialization if it fails
- WiFi connection attempts timeout after 30 seconds
- MQTT reconnection is attempted automatically if connection is lost

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Author

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- Adafruit for OLED and INA226 libraries
- Arduino for ESP32 support
- MQTT for communication
- PID controller library
- INA226 datasheet for reference

Future Improvements

- Implement a web interface for configuration
- Add temperature monitoring
- Optimize PID parameters
- Improve error handling and logging
- Add user-friendly configuration options
- Add safety features like overcurrent protection
- Implement overvoltage and undervoltage protection
- Add a buzzer for alerts

Contributors

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